Verification
04_H_Simulator_FVM_1D_Compressible_Heterogeneous

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1 Objective

The purpose of this verification exercise is to demonstrate that the numerical simulator correctly reproduces the analytical solution for one-dimensional slightly compressible single-phase flow in a heterogeneous porous medium.

2 Governing Equation

For one-dimensional slightly compressible single-phase flow in a homogeneous porous medium, conservation of mass combined with Darcy's law gives

$$\frac{\partial(\phi\rho)}{\partial t} = \frac{\partial}{\partial x}\left(\frac{\rho k\,(x)}{\mu}\frac{\partial P}{\partial x}\right), \tag{1}$$

where

- P is pressure [Pa],
- ρ is fluid density [kg/m^3],
- ϕ is porosity [-],
- $k\,(x)$ is permeability [m^2],
- μ is fluid viscosity [Pa·s].

For a slightly compressible fluid,

$$c_f = \frac{1}{\rho}\frac{\partial\rho}{\partial P}, \tag{2}$$

and assuming constant porosity and compressibility, the density is given by

$$\rho = \rho_0 e^{c_f(P-P_0)}, \tag{3}$$

where ρ_0 is the reference density at pressure P_0. For small pressure variations, the density may be linearized as

$$\rho \approx \rho_0\left[1 + c_f(P-P_0)\right], \tag{4}$$

which leads to the pressure diffusivity equation,

$$\phi c_f \frac{\partial P}{\partial t} = \frac{\partial}{\partial x}\left(\frac{k(x)}{\mu}\frac{\partial P}{\partial x}\right). \tag{5}$$

The domain is defined on

$$0 \leq x \leq L_x, \tag{6}$$

with fixed-pressure boundary conditions

$$P(0,t) = P_L, \qquad P(L_x,t) = P_R, \tag{7}$$

and the initial condition

$$P(x,0) = P_i. \tag{8}$$

3 Analytical Solution

To validate the numerical simulator, a steady-state solution is used. In steady state, the flux is constant such that,

$$\frac{d}{dx}\left(k\frac{dP}{dx}\right) = 0. \tag{9}$$

As the flux is constant everywhere, it follows that the Darcy fluxes of the two sections of permeability are equivalent,

$$q = -\frac{k}{\mu}\frac{dP}{dx}. \tag{10}$$

The pressure gradients are therefore different and equal to

$$\frac{dP}{dx} = -\frac{\mu q}{k} \tag{11}$$

in each section. In the first layer, therefore,

$$P(x<a) = P_L - \frac{\mu q}{k_1}x \tag{12}$$

and in the second layer,

$$P(x>a) = P(a) - \frac{\mu q}{k_1}(x-a) \tag{13}$$

where a is the permeability change location with,

$$P(a) = P_L - \frac{\mu q}{k_1}a. \tag{14}$$

Then, solving for q yields,

$$q = \frac{P_L - P_R}{\mu\left(\frac{a}{k_1} + \frac{L_x - a}{k_2}\right)}. \tag{15}$$

By substituting in q,

$$P(x) = \begin{cases} P_L - \dfrac{P_L - P_R}{\frac{a}{k_1} + \frac{L_x - a}{k_2}}\frac{x}{k_1}, & 0 \leq x \leq a, \\ P_L - \dfrac{P_L - P_R}{\frac{a}{k_1} + \frac{L_x - a}{k_2}}\left(\frac{a}{k_1} + \frac{x-a}{k_2}\right), & a \leq x \leq L_x. \end{cases} \tag{16}$$

4 Simulation Parameters

The simulation parameters are given in Table 1.

Parameter	Symbol	Value
Number of cells	N_x	10000
Domain length	L_x	10 m
First permeability domain	a	2 m
Permeability	k_1	1×10^{-13} m^2
Permeability	k_2	1×10^{-12} m^2
Viscosity	μ	0.1 Pa·s
Left pressure	P_L	10×10^6 Pa
Right pressure	P_R	1×10^5 Pa
Compressibility	c_f	5×10^{-8} Pa^{-1}
Reference density	ρ_0	1000 kg/m^3
Reference pressure	P_0	1×10^5 Pa

Table 1: Simulation parameters.

5 Results

5.1 Pressure Distribution

The pressure solutions for both the MATLAB simulator, Figure 1, and the Python simulator, Figure 2, match well.

6 Discussion

The numerical pressure distribution reproduces the analytical linear pressure solution at steady state.

7 Conclusion

The simulator successfully reproduces the analytical solution for one-dimensional slightly compressible flow in a heterogeneous porous medium. This provides verification that the finite volume implementation is correctly solving the governing equations for this benchmark problem.

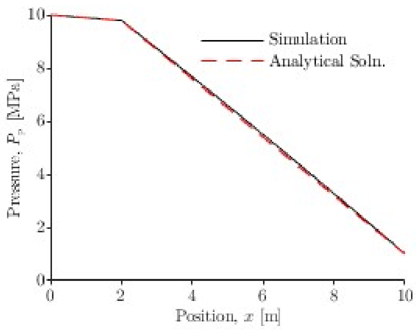

Figure 1: Comparison between analytical and numerical pressure solutions for the MATLAB simulator.

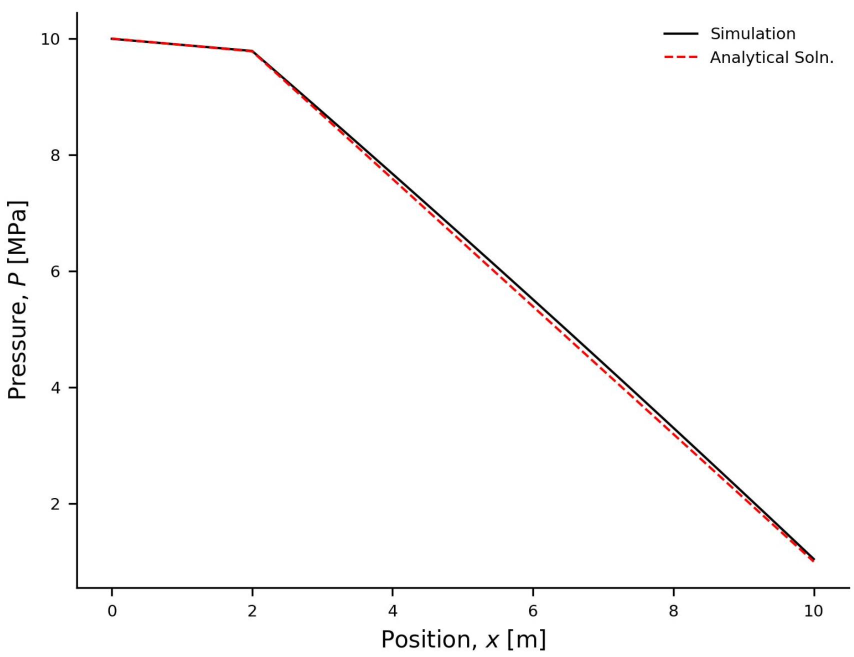

Figure 2: Comparison between analytical and numerical pressure solutions for the Python simulator.